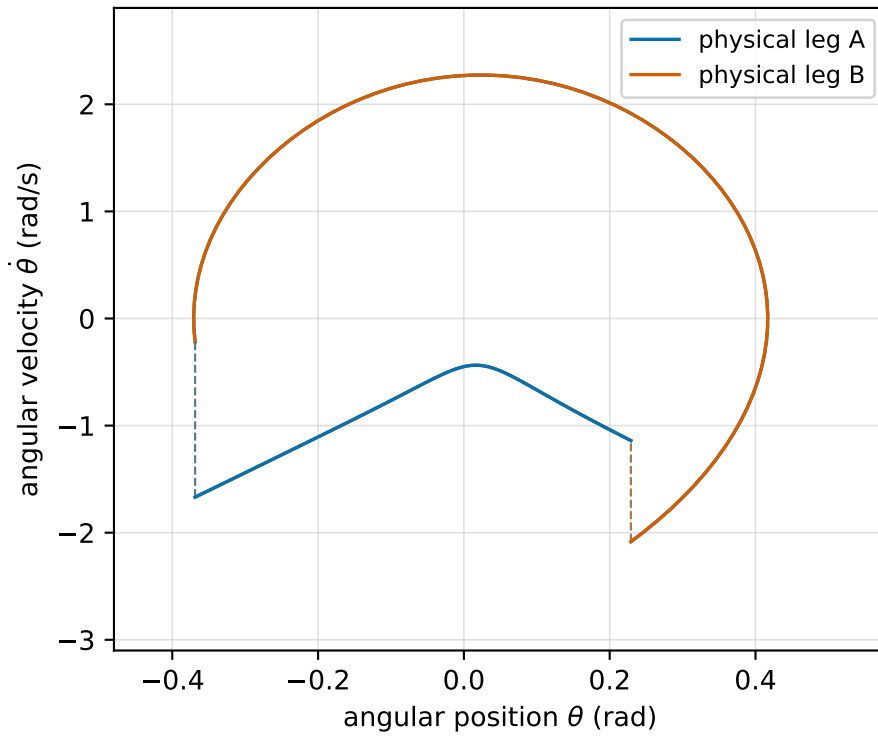
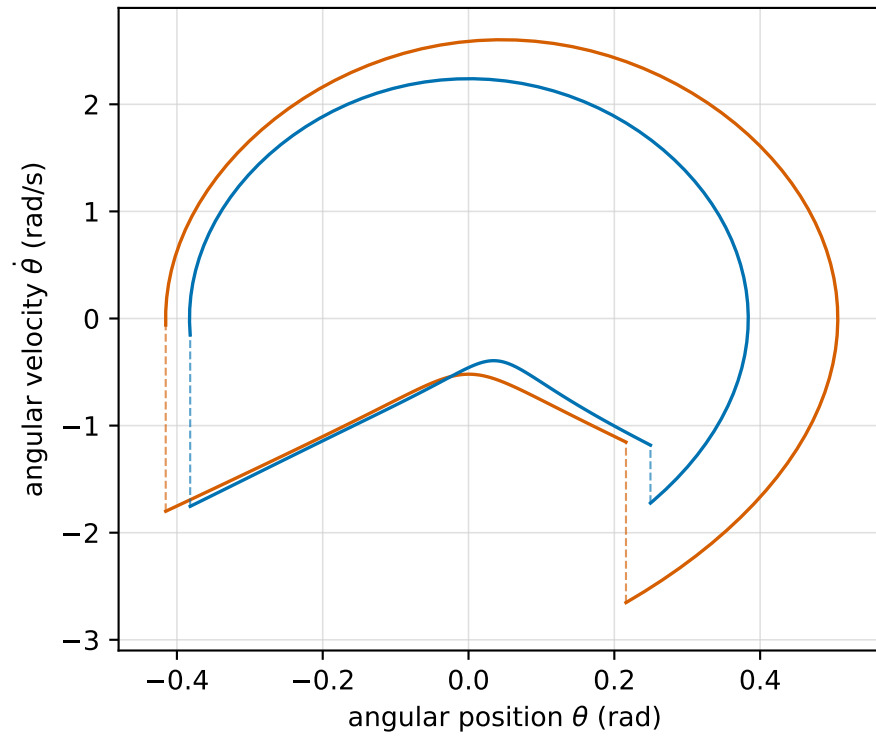


Passive Compass Gait: Phase-Plane Period Doubling

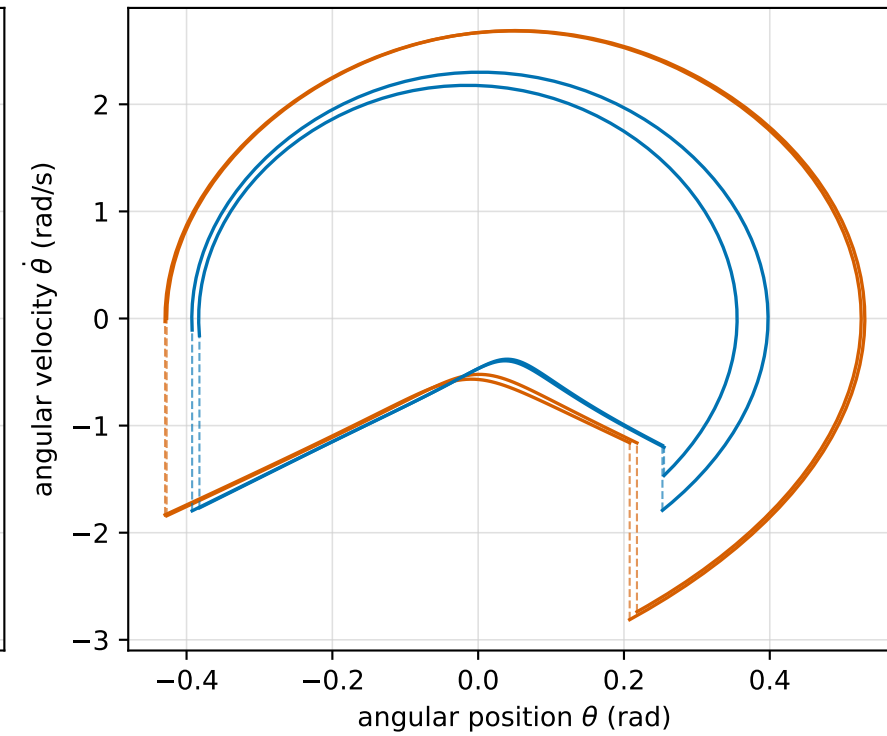
(a) Period 1 | $\phi = 4.00^\circ$



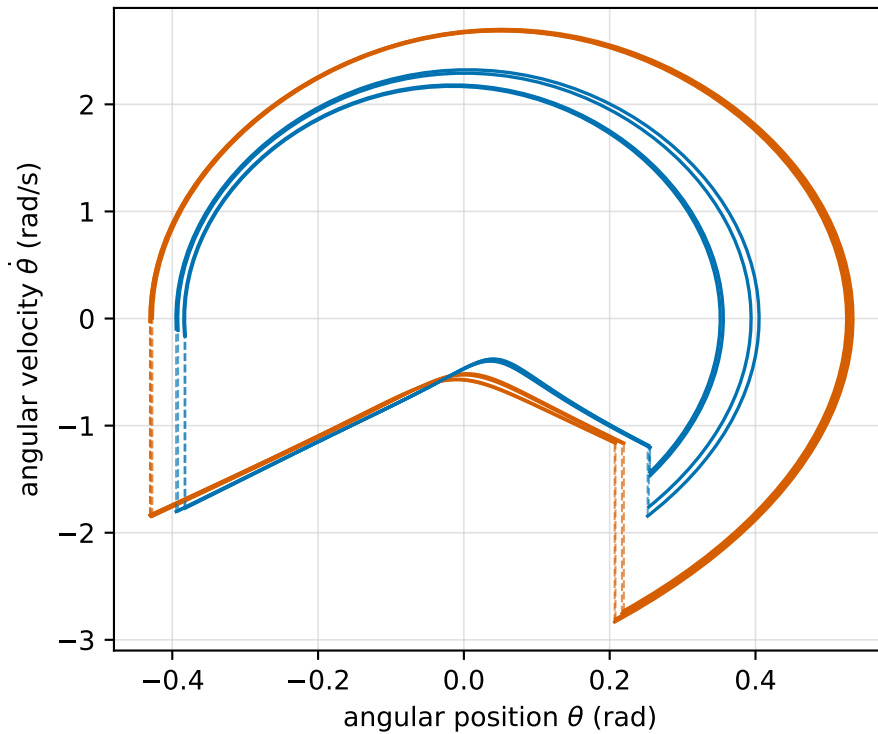
(b) Period 2 | $\phi = 4.75^\circ$



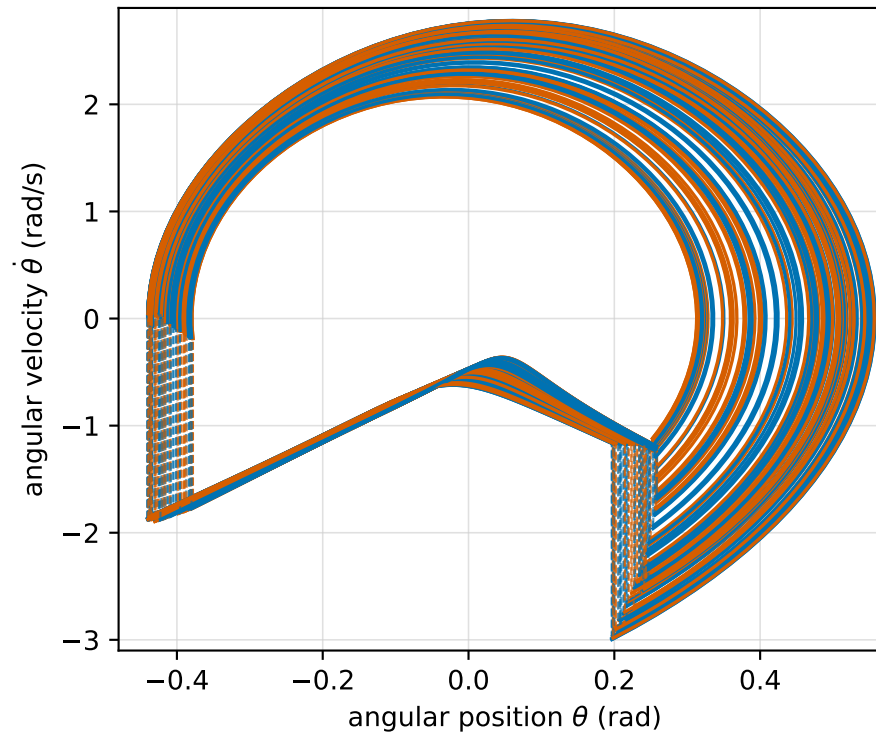
(c) Period 4 | $\phi = 5.00^\circ$



(d) Period 8 | $\phi = 5.02^\circ$



(e) Chaotic gait | $\phi = 5.20^\circ$



Compass-gait period-doubling cascade

Nominal model: $\mu = 2$, $\beta = 1$, $l = 1$ m

Solid curves: continuous swing dynamics
Dashed curves: instantaneous impact reset

Colors follow physical legs across the stance/nonstance label swap.